

Course: Individual Software Development Process' 2026

Iteration Report: Iteration 2

By ISANPOWER888 (ISP888)

Chosen Irl Challenge: Project B: Lab Workflow & Request Management



Theewasu Aekthong	6810545701
Kantee Laibuddee	6710545440
Tanon Likhittaphong	6710545547
Inthat Niramarn	6810545999

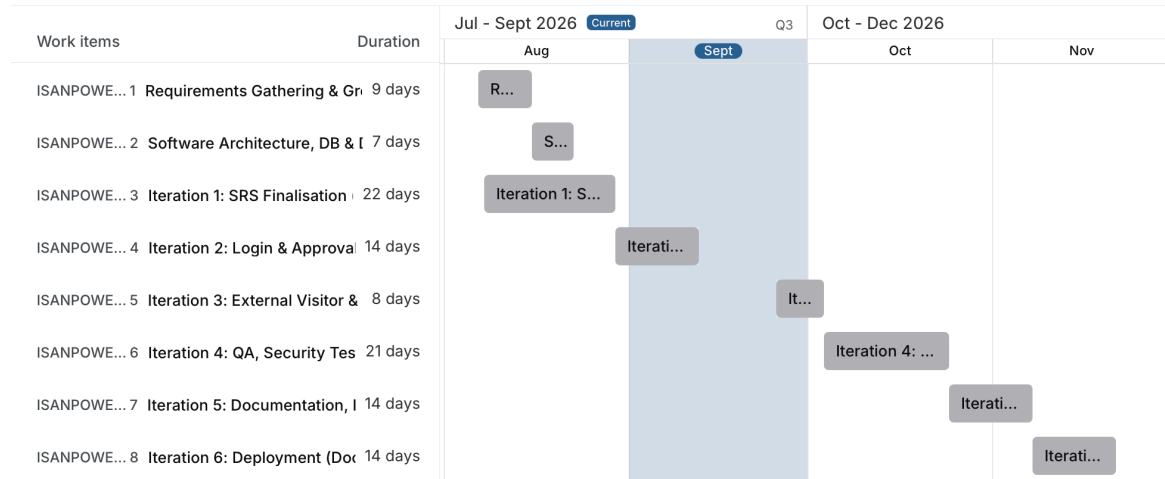
Table of Contents

1. Project Snapshot.....	1
1.1 Project-Wide Gantt Chart.....	1
1.2 Current Iteration Gantt Chart.....	1
2. Sprint/Board Status Snapshot.....	2
3. Scope Addition/Change Log.....	3
4. Retrospective.....	3
5. Testing & Quality Notes.....	4
6. Project Risk Register.....	4
7. Individual Contribution Log.....	5

1. Project Snapshot

Project Repository: <https://github.com/lnthat6810545999/ISANPOWER888>

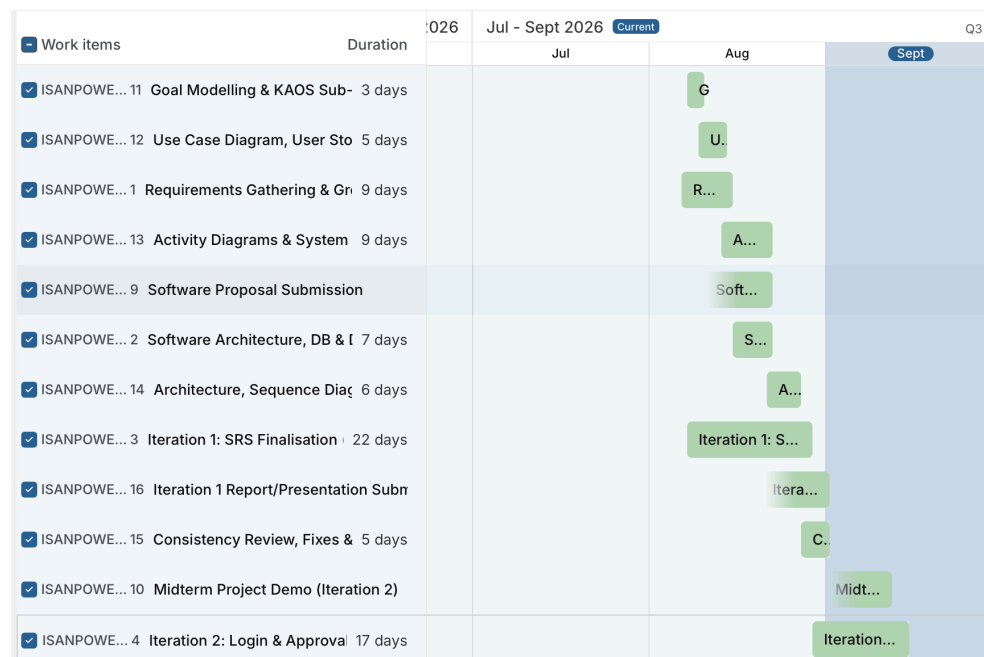
1.1 Project-Wide Gantt Chart



Critical Path: Requirements sign-off → Architecture/DB finalisation → SRS Iteration 1 submission (US-1,2,3) → Login & Approval workflow (Iteration 2, Midterm Demo) → External Visitor module (Iteration 3) → QA & Code Quality pass (Iteration 4) → Documentation (Iteration 5) → Deployment (Iteration 6)

Critical Path Duration: 106 calendar days (Aug 10 – Nov 24)

1.2 Current Iteration Gantt Chart



Critical Path: Goal modelling → Use case & user stories → Activity diagrams & SRS → Architecture & sequence diagrams → Consistency review & report submission → Midterm Presentation → Iteration 2

Critical Path Duration: 39 calendar days (Aug 11 – Sep 18)

2. Sprint/Board Status Snapshot

Iteration Goal: Implement and integrate the approval-gated request workflow committed for Iteration 2 TA self-assignment/status updates (US-4) and Lab Manager approve/reject authorisation (US-5) together with role-based authentication (US-9), converting the Iteration 1 SRS foundation into a working system where TA claim is blocked until an authorised Manager decision is recorded and every user logs in with role-scoped access, directly supporting SG-3 (Assign/Update Request Status) and the approval-authorisation requirements traced to URS-4/URS-5.

Capacity Declared: 3 user stories (US-4, US-5, US-9).

User Story	Description	Owner	Status	Within Capacity (Y/N)	Notes
US-4	As a TA, I want to assign a request to myself or update its status so that progress is tracked.	Theewasu	Done	Y	TA self-assign (claim) now gated by manager approval; TaQueue.tsx, labRequestController.ts updated
US-5	As a lab manager, I want to approve or reject requests that need authorisation so that resource use stays controlled.	Inthat, Theewasu	Done	Y	Theewasu implemented the approve/reject request logic and the approval-first gate ; Inthat built the underlying Lab Manager backend session/role verification and membership routes that the approval capability

					runs.
US-9	As a registered user, I want to log in to the system so that I can access functions according to my role.	Kantee	Done	Y	Google Sign-In added alongside existing password login

3. Scope Addition/Change Log

Note: Log any item added, expanded, or swapped in after planning began. Empty is a good sign; frequent entries signal scope creep.

Item added/changed	Reason	What as dropped to compensate (one-in-one-out)
Admin role added mid-sprint (commit 8523931), then removed same sprint (commit 058560b)	Added to support account bootstrap, then judged redundant once Lab Manager "Approve & assign role" membership flow covered the same need	Admin role removed entirely replaced one-for-one by manager-driven membership approval
Google sign-in + guest/pending-membership flow	Needed as a real login path beyond seeded password accounts, ahead of presentation	None dropped

4. Retrospective

What went well	What didn't go well	Action items for next sprint
The workflow has become significantly clearer, with the system now visibly structured into distinct, well-defined stages	The Admin role was added mid-sprint, then removed once the Lab Manager membership-approval flow was found to cover the same need — an avoidable scope churn.	Confirm role/permission model with the team before implementation, not after.

5. Testing & Quality Notes

- CI workflow broke and was repaired in the same commit that removed the Admin role ("fix: remove admin role and repair CI workflow").
- New/updated tests this sprint: googleAuthRoutes.test.ts, membershipRoutes.test.ts, retiredAdminRoutes.test.ts (new); labRequestRoutes.test.ts (147 lines changed to cover the new approval-gate logic); apps/web/scripts/request-status.test.mjs updated.
- Frontend/backend/Prisma enum consistency is checked via `npm run test:contracts`. - Standard gate before merge: lint, typecheck, build (both apps) + API integration tests against a dedicated `\_test` database.

6. Project Risk Register

Risk ID	Description	Likelihood (L/M/H)	Impact (L/M/H)	Mitigation/Contingency	Status
R-01	Migration 20260920000000_approval_first_workflow resets ALL open requests (even previously approved/in-progress) back to pending/submitted and clears assignees	M	H	Communicate to whole team before running; back up DB first; never run `prisma migrate reset`; re-review/re-claim affected requests before any demo	Monitoring
R-02	Google OAuth credentials aren't in the repo Google login demo could fail on a teammate's machine if `.env` isn't configured	M	M	Document Google Cloud Console setup step-by-step; keep seeded password accounts as a fallback login path for presentation	Open

R-03	Admin role was added then removed within the same sprint teammates who already applied the add-Admin migration must also apply the removal migration or their local DB state diverges	L-M	M	All teammates run `npx prisma migrate deploy` after pulling `main`; do not reset local databases	Monitoring
------	---	-----	---	--	------------

7. Individual Contribution Log

Student Name	Tasks owned this sprint	Course concept(s) applied	Evidence	Self-rated effort (1-5)
Theewasu Aekthong	Approval-workflow (US-4)/ report	SRS, Activity Diagram, SQD	Commit 8530834	5
Kantee Laibuddee	Google-login (US-9), remove admin-role	URS, Activity Diagram, SQD	Commits 4d33996, c9b37f1, 058560b	5
Tanon Likhittaphong	Slide / Update-web-logo	Requirement Engineering(5th-step)	Commit 8733bd0	5
Inthat Niramarn	lab-manager-controller (US-5 backend)	SRS, SA, SQD	Commit 8523931	5